

1. Simplify the following expressions using Boolean algebra

a) $A+AB$
 $A(1+B)$
 $A(1)$
 $A(1)=A$

b) $AB+AB'$
 $A(B+B')$
 $A(1)$
 $A(1)=A$

c) $A'BC+AC$
 $C(A'B+A)$
 $C(A+B)$
 $C(A+B)$

d) $A'B+AB'C+ABC$
 $A'B+AB(C'+C)$
 $A'B+AB(1)$
 $A'B+AB$
 $B(A'+A)$
 $B(1)$
 B

2. Using DeMorgan's theorem, show that:

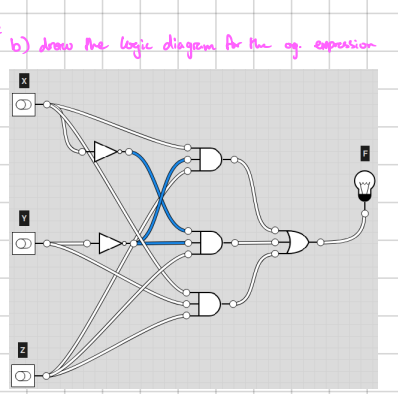
a) $(A+B)(A'+B')=0$
 $(A+B)(A+B)'=0$
 $A+B=0$
 $A=0, B=0$
 $0+0=0$
 $0=0$

b) $A+A'B=A$
 $A+A'(B+B')=A$
 $A+A'(1)=A+A'$
 $A+A'=1$
 $1=1$

3. Given the Boolean function: $F = xy'z + x'y'z + xyz$

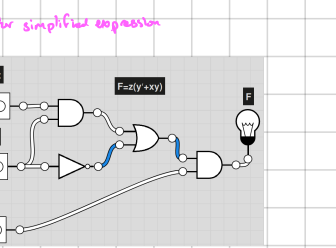
a) Draw the truth table of the function

x	y	z	y'	xy'z	x'y'z	xyz	F
0	0	0	1	0	0	0	0
0	0	1	1	0	1	0	1
0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	0
1	0	0	1	0	0	0	0
1	0	1	1	1	0	0	1
1	1	0	0	0	0	0	0
1	1	1	0	0	0	1	1



d) Show truth table for simplified expression

x	y	z	y'	xy	y'+xy	F = z(y'+xy)
0	0	0	1	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	1	0	1	0
1	0	1	1	0	1	1
1	1	0	0	1	1	0
1	1	1	0	1	1	1



c) Simplify the expression

$$F = xy'z(x+y) + xyz = y'zC + xyz$$

$$\Rightarrow F = y'z + xyz \Rightarrow z(y' + xy)$$

$$\Rightarrow F = z(y' + xy)$$

4. Simplify the following Boolean functions

a) $F = AB + BC + B'C$
 $F = AB + C(B+B') = AB + C(1) = AB + C$

b) $F = A + A'B$
 Already simplified:
 $F = A + A'B$

c) $F = A'B'C + A'BC + AB'$
 $F = A'C(B'+B) + AB' = A'C(1) + AB' = A'C + AB'$

d) $F = AB + (AC)' + AB'C(AB+C)$
 $F = AB + A'C + AB'(CAB + ABC) = AB + A'C + AB'ABC = AB + A'C + AB'C$

5. Obtain the sum of products form of the following functions using Miniterms:

a) $F(A,B,C) = A + BC$
 $A \Rightarrow A(B+B')(C+C') = ABC + AB'C + AB'C' + ABC'$
 $BC \Rightarrow BC(A+A') = ABC + A'BC$
 $F = ABC + AB'C + AB'C' + ABC' + ABC + A'BC = ABC + AB'C + AB'C' + A'BC$

b) $F(A,B,C,D) = AB + ACD$
 $AB \Rightarrow AB(C+C')(D+D') = ABCD + ABC'D + AB'C'D + ABC'D'$
 $ACD \Rightarrow ACD(B+B') = ABCD + A'BCD$
 $F = ABCD + ABC'D + AB'C'D + ABC'D' + ABCD + A'BCD = ABCD + ABC'D + AB'C'D + A'BCD$

6. Obtain the product of sums form of the following functions

a) $F(A,B,C) = A + BC$
 $F' = A'(B+C)' \Rightarrow F' = (A'+B')(A'+C)' = (A'+B')(A'+C')$

7. Find the complement of each of the functions by applying DeMorgan's theorem:

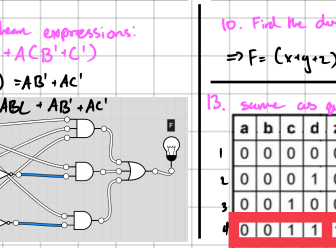
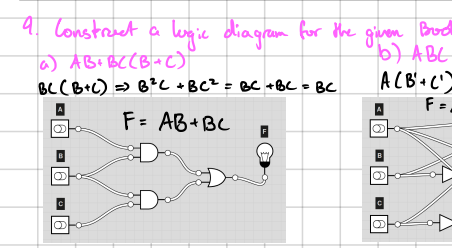
a) $F1 = x'y'z' + x'y'z$
 $F1' = (x+y+z)(x+y+z')$

b) $F2 = x(x'z' + yz)$
 $F2' = x'(x+z)(y'+z')$

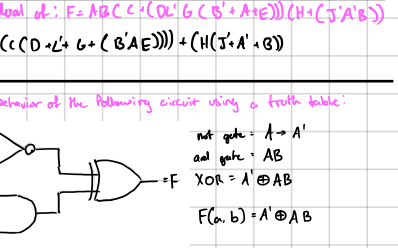
8. Simplify the following expression using Boolean algebra

a) $A'BC + ABC = AC(B+B') = AC$
 $A'BC + ABC = BC(A'+A) = BC$
 $A'BC + ABC = AC(B+B') = AC$
 $A'BC + ABC = AC(B+B') = AC$

11. Find the dual of: $F = ABC + C'(D'E' + A'E))((H + J'A'B'))$
 $\Rightarrow F' = (A+B)(C(D'+E'+(A'+E')))((H'+J'A'B'))$



10. Find the dual of the function: $F = xyz + x'y'z + y'z$
 $\Rightarrow F' = (x+y+z)(x'+y'+z)(y'+z)$



12. Implement the function of the following truth table using sum-of-products form:

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

rows with output 1:
 4: $A=0, B=1, C=1 \Rightarrow A'B'C$
 6: $A=1, B=0, C=1 \Rightarrow AB'C$
 7: $A=1, B=1, C=0 \Rightarrow ABC'$
 8: $A=1, B=1, C=1 \Rightarrow ABC$

SOP form:
 $F(A,B,C) = (A'B'C) + (AB'C) + (ABC') + (ABC)$

13. Answer question 12

a	b	c	d	z
1	0	0	0	0
2	0	0	1	0
3	0	1	0	0
4	0	1	1	1
5	1	0	0	0
6	1	0	1	0
7	1	1	0	0
8	1	1	1	1
9	1	0	0	0
10	1	0	1	0
11	1	0	0	1
12	1	0	1	1
13	1	1	0	0
14	1	1	0	1
15	1	1	1	0
16	1	1	1	1

rows with output(z) > 1:
 4: $a'b'cd$
 8: $a'b'cd$
 12: $a'b'cd$
 16: $abcd$

SOP form:
 $z(a,b,c,d) = (a'b'cd) + (a'b'cd) + (a'b'cd) + (abcd)$

15. Show the behavior of the following circuit using a truth table:

A	B	C	F(A,B,C)
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

A	B	A'	AB	A'⊕AB
0	0	1	0	1
0	1	1	0	1
1	0	0	0	0
1	1	0	1	1